

 **Live Web Application:** <https://roamcraft.onrender.com>
 **GitHub Repository:** <https://github.com/sagun-glitch/RoamCraft>

1. Executive Summary

RoamCraft is a lightweight, AI-driven vacation planning platform built with **Node.js, Express, and Google Gemini 2.5 Flash**. It generates comprehensive, day-by-day travel itineraries delivered via **Server-Sent Events (SSE)** real-time streaming based on user destination, duration, budget tier, and custom activity preferences.

2. Key Highlights & Features

- Real-Time AI Streaming:** Delivers instant markdown-formatted itinerary updates chunk-by-chunk using Server-Sent Events (SSE) without long loading delays.
- Precision Personalization:** Tailors daily itineraries based on budget tier (Budget-friendly, Moderate, Luxury) and custom interest preferences (e.g. food tours, historical places, beaches, hiking).
- Interactive Day-by-Day Dashboard:** Renders structured day blocks with morning/afternoon/evening schedules, interactive packing checkboxes, and local travel tips.
- One-Click Export & Printing:** Integrated print/PDF engine formats custom itineraries into clean, styled documents.
- Scandinavian Terracotta Sand Theme:** High-contrast, clean UI designed for maximum readability without gradients or glassmorphism.

3. Technical Stack & Architecture

Layer	Technology	Details
Live Hosting	Render Cloud Platform	Deployed live at https://roamcraft.onrender.com
Backend Engine	Node.js / Express.js	Manages REST endpoints, static file serving, and SSE streaming pipeline

Layer	Technology	Details
Generative AI	Google Gemini 2.5 Flash	High-speed generative AI model configured for structured streaming output
Frontend UI	HTML5 / Vanilla CSS / ES6+ JS	Dashboard interface built with Space Grotesk & IBM Plex fonts, and Lucide Icons
Streaming Protocol	Server-Sent Events (SSE)	HTTP streaming protocol for progressive chunk rendering in client browser

4. API Endpoint Reference

POST `/api/generate-itinerary`

Request Payload (JSON):

```
{
  "destination": "Rome, Italy",
  "days": 3,
  "budget": "Moderate",
  "interests": "historic ruins, authentic pasta, neighborhood walks"
}
```

5. Local Development & Deployment

- Clone repository: `git clone https://github.com/sagun-glitch/RoamCraft.git`
- Install dependencies: `npm install`
- Configure environment file `.env` with `GEMINI_API_KEY=YOUR_KEY`
- Start server: `npm start` (Open `http://localhost:3000` or visit `https://roamcraft.onrender.com`)

6. Metadata & Credits

- Project Name:** RoamCraft
- Author / Owner:** Sagun Mardi (`sagun-glitch`)
- Live Application:** `https://roamcraft.onrender.com`
- Repository:** `https://github.com/sagun-glitch/RoamCraft`